

Business Intelligence: Multidimensional Data Analysis

Business Intelligence (BI) requires analyzing and interpreting data to support decision-making in organizations. *Data Warehousing* is the set of actions comprising collecting, storing, and managing large volumes of data from various sources for *BI* purposes. We will explore some tools in BI for data integration, querying, and multidimensional analysis.

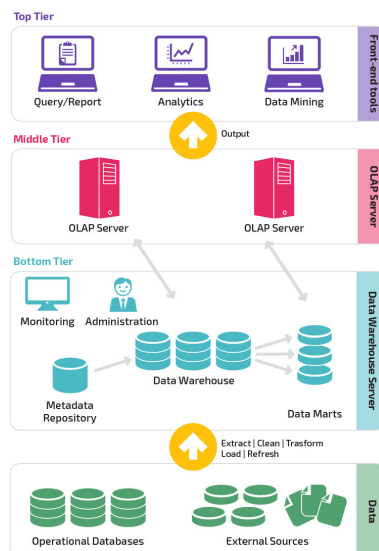
Objectives:

- 1- Explore some *BI and Data Warehousing* tools and techniques.
- 2- Explore some basics for efficient data storage and management in *BI* scenarios.
- 3- Explore tools for implementing *OLAP (Online Analytical Processing)* and its concepts.

1 Useful Notions & Syntaxes

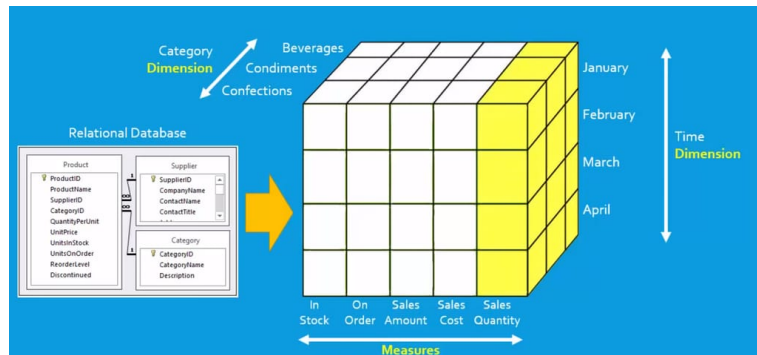
(a) Business Intelligence & Data Warehouse

- i) *BI* is a process for analyzing data and inferring insights to help businesses make decisions. In an effective *BI* process, analysts and data scientists discover meaningful assumptions and can answer them using available data.
- ii) A *data warehouse* is a *relational* database that gathers structured data from across an entire organization. It pulls together data from multiple sources. The *data warehouse selects, organizes and aggregates data for efficient comparison and analysis.*



(b) BI & OLAP Tools

- i) *BI* relies on efficient tools for storing, managing, and analyzing large datasets. In this lab, we will work with *PostgreSQL* for database management, *Mondrian* as an *OLAP* engine, and *JRubik* as a graphical interface for multidimensional analysis. These tools together allow us to build, explore, and query data *cubes* to support powerful decision-making through *OLAP (Online Analytical Processing) systems*.
- ii) *OLAP(Online Analytical Processing)* allows users to quickly analyze data from multiple *perspectives (dimensions)*, making it easier to discover patterns, trends, and insights. It typically operates on data *cubes* built from relational databases like PostgreSQL. *OLAP systems* are capable of combining classic tables in a sort of table of tables, *which can be visualized as a 3D OLAP Cube for simplicity*.



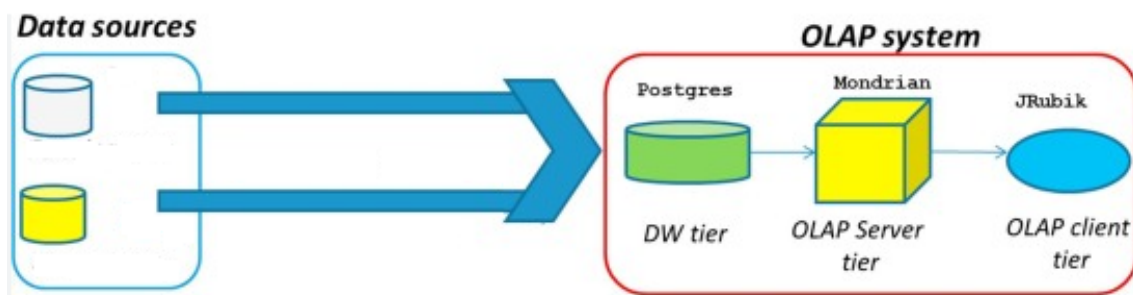
iii) Some Tools Overview:

- 1.) *PostgreSQL*: An open-source, powerful relational database system that stores and organizes data for analysis.
- 2.) *Mondrian*: An open-source OLAP engine that allows users to perform fast, multidimensional queries (like slicing and dicing data cubes) directly on relational databases.
- 3.) *JRubik*: A graphical user interface tool that connects to Mondrian OLAP cubes, helping users easily visualize and interact with multidimensional data without writing complex queries.

How They Work Together:

- 1- *PostgreSQL* stores the raw, relational data (tables from *imijdbManager*: Staff, Show, Rating,...).
- 2- *Mondrian* connects to *PostgreSQL* and reads the data to build *OLAP cubes*.
- 3- *JRubik* connects to *Mondrian* to visualize and explore these *cubes* easily, without manually writing OLAP queries.

Together, they form a complete BI solution that supports flexible, fast, multidimensional analysis.

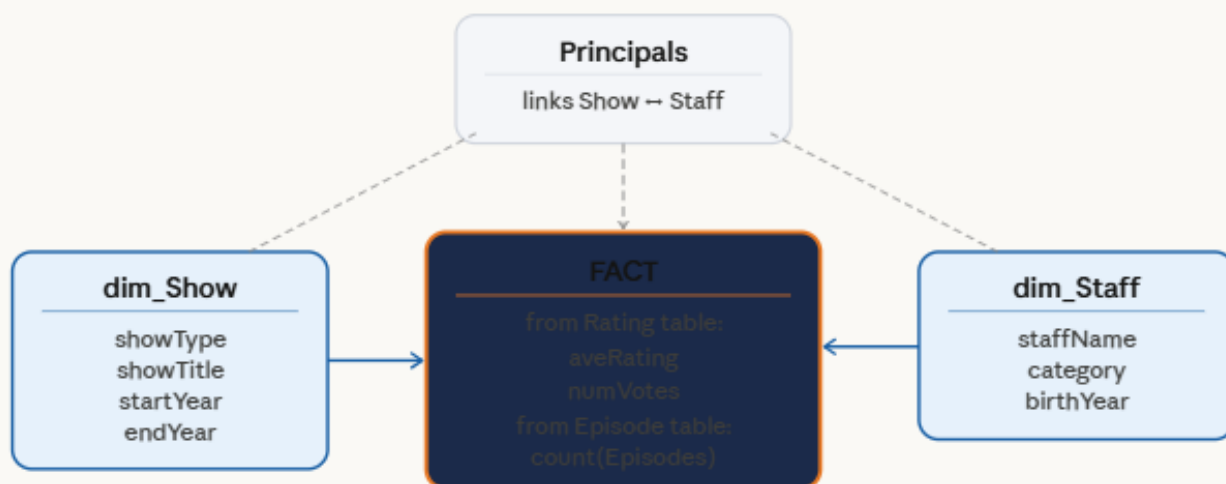


② Lab Work

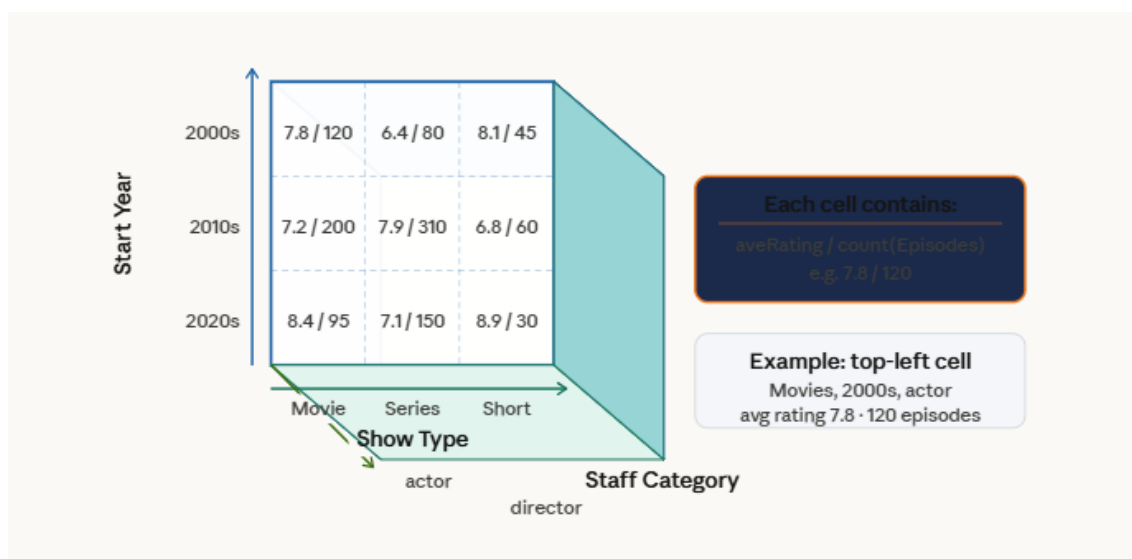
The work for this lab is more like exploration, using the *IMDb* (*Internet Movie Database*), from the previous labs.

i) Explore Dimensions and Measures

- In the JRubik interface, you'll have a list of dimensions (e.g., Show, Staff, ...) and measures (e.g., Rating, Episode, ...).

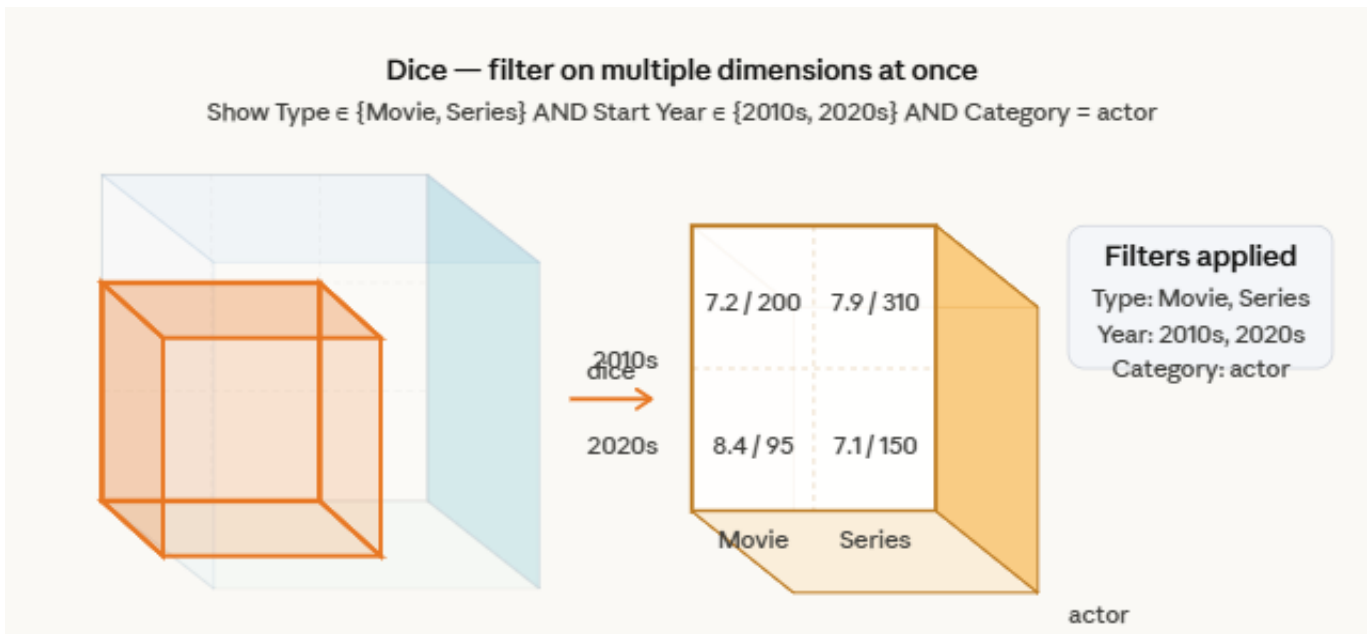
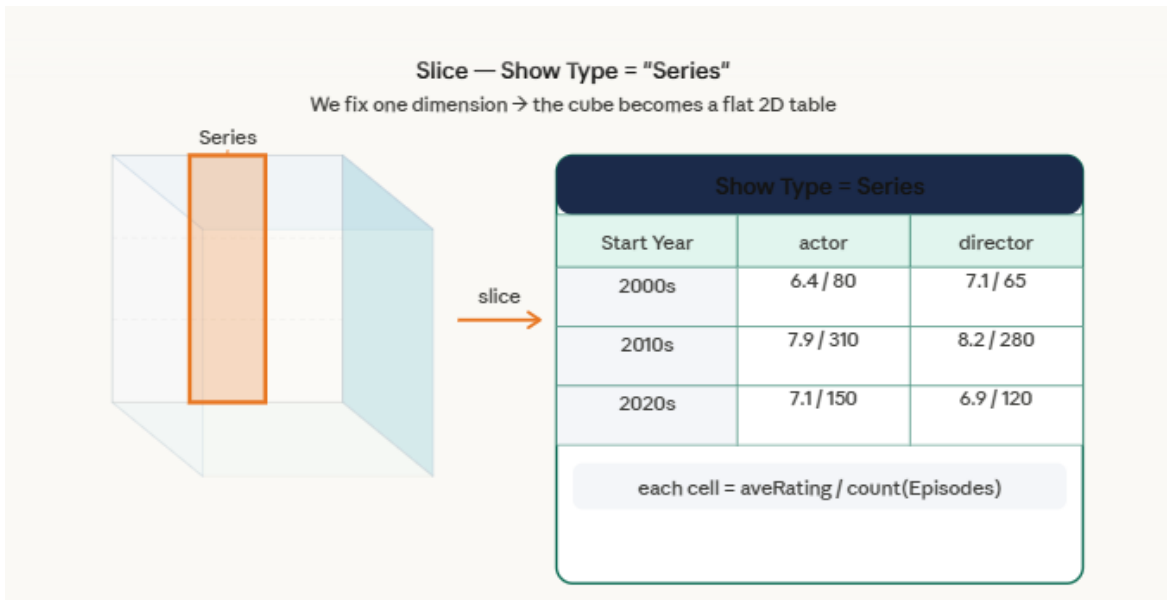


Here's what makes sense as an OLAP cube from *IMDb*:

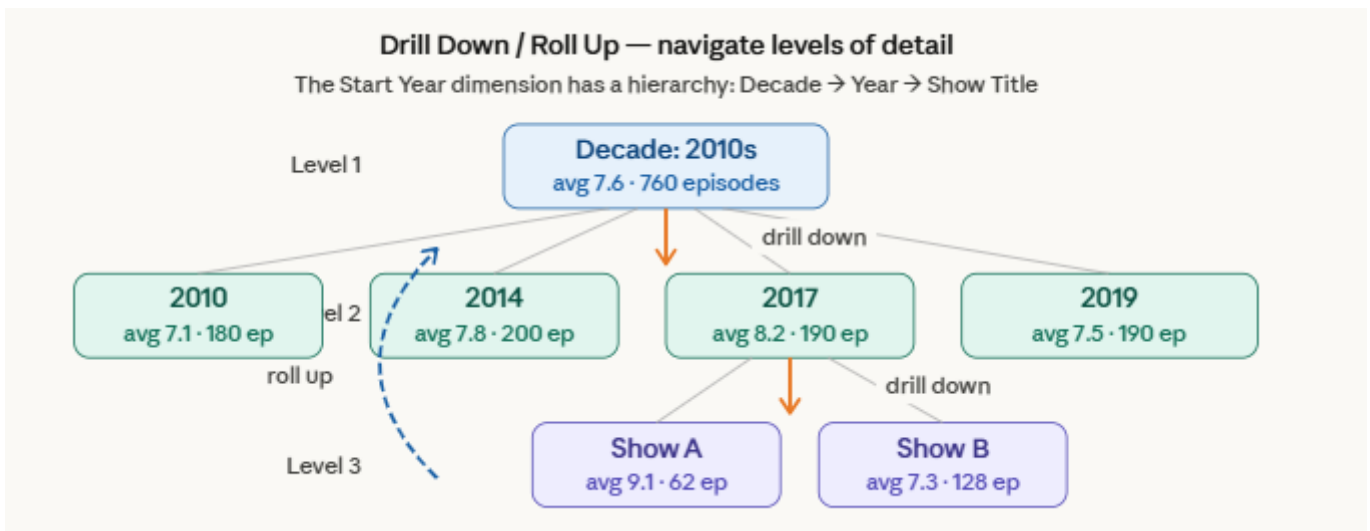


-You can interact with these elements to analyze your data:

Slice and Dice: Filter and rearrange data to view it from different perspectives.



Drill Down/Up: Navigate through hierarchical levels of data for detailed analysis.



Pivot: Rotate data axes to view different summaries.

Pivot — swap axes, same data, new perspective
Rows → Columns: Start Year and Show Type switch places

Before rows = Start Year					After rows = Show Type			
Year \ Type	Movie	Series	Short		Type \ Year	2000s	2010s	2020s
2000s	7.8 120 ep	6.4 80 ep	8.1 45 ep		Movie	7.8 120 ep	7.2 200 ep	8.4 95 ep
2010s	7.2 200 ep	7.9 310 ep	6.8 60 ep		Series	6.4 80 ep	7.9 310 ep	7.1 150 ep
2020s	8.4 95 ep	7.1 150 ep	8.9 30 ep		Short	8.1 45 ep	6.8 60 ep	8.9 30 ep

Answer the following questions: What do you think?

1.) **Table Before** : For each year, how did each show type perform?

2.) **Table After** : For each show type, how did it perform across years?

ii) **Generate Reports**

- We can use JRubik's reporting features to create visual representations of your data.
- We can customize the layout and format as needed to highlight key insights.